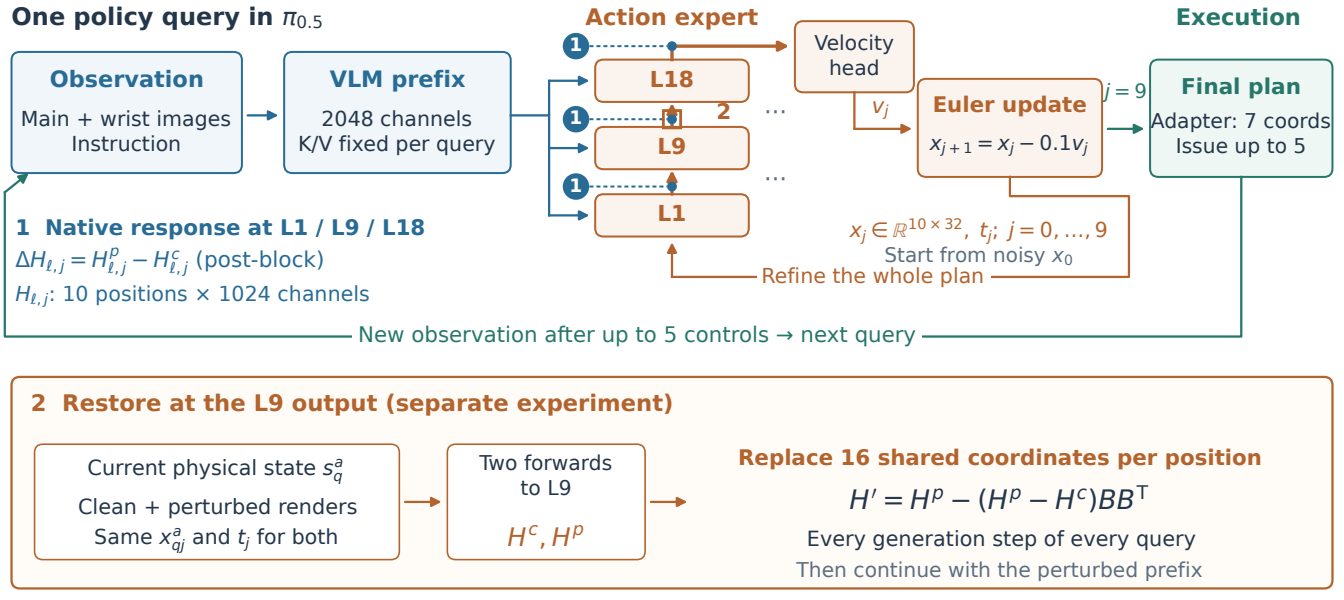
One policy query in $\pi_{0.5}$
Observation
Main + wrist images
Instruction
VLM prefix
2048 channels
K/V fixed per query
Action expert
1
L18
1
2
L9
1
L1
Velocity head
v_j
Euler update
$x_{j+1} = x_j - 0.1 v_j$
$j = 9$
Execution
Final plan
Adapter: 7 coords
Issue up to 5
1 Native response at L1 / L9 / L18
$\Delta H_{\ell,j} = H^p_{\ell,j} - H^c_{\ell,j}$ (post-block)
$H_{\ell,j}$: 10 positions × 1024 channels
$x_j \in \mathbb{R}^{10 \times 32}$, t_j; $j = 0, \ldots, 9$
Start from noisy x_0
Refine the whole plan
New observation after up to 5 controls → next query
2 Restore at the L9 output (separate experiment)
Current physical state s^a_q
Clean + perturbed renders
Same x^a_{qj} and t_j for both
Two forwards to L9
H^c, H^p
Replace 16 shared coordinates per position
$H' = H^p - (H^p - H^c)BB^\top$
Every generation step of every query
Then continue with the perturbed prefix